

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Compute the polar representation of $2 + 3I$.

Verify that the polar representation corresponds to $2 + 3I$.

2. Does $p = x^2 + 2x + 3$ factor over $\mathbb{Q}[x]$?

Extend $\mathbb{Q}$ with a formal root of p .

What is the other root of p in the extended field?